The Impact of Green Economy Practices on Firm Innovation: Evidence from Indonesian Manufacturing and Service Sectors (2023)

Kornelius Hasiholan Pardosi, Nathaniel Toga Siringo Ringo, William Philip Karnadi

Faculty of Economics and Business, University of Indonesia

korneliuspardosi@gmail.com, nathanieltogasiringoringo@gmail.com, william.philipk23@gmail.com

ABSTRACT

This study investigates whether Green Economy Practice (GEP) adoption increases the likelihood of product and process innovation among Indonesian firms using cross-sectional firm-level data from the 2023 World Bank Enterprise Survey (WBES), covering 794 firms across manufacturing, retail, and other services sectors. Grounded in the Porter Hypothesis, Resource-Based View, and Institutional Theory, binary logistic regression with survey-weighted heteroskedasticity-consistent standard errors and Average Marginal Effects (AME) were employed, with probit estimation for robustness. Results reveal that GEP adoption has no statistically significant effect on product innovation, suggesting that translating green capabilities into new commercial offerings requires broader enabling conditions, including R&D infrastructure and university-industry linkages, largely absent in Indonesia's current industrial ecosystem. In contrast, each additional green practice adopted raises the probability of process innovation by approximately 7.46 percentage points against a baseline mean of just 2.14%, an effect concentrated among smaller and older firms, and confirmed consistently across both logit and probit specifications. With the average Indonesian firm having adopted only roughly two of ten possible green practices and a majority adopting none, Indonesia's green innovation transition remains deeply nascent, necessitating credible regulatory enforcement, targeted financial incentives for SMEs, and demand-side green market development to unlock the full innovation dividend.

Keywords: Green Economy Practices, Firm-Level Innovation, Process Innovation, SMEs, Green Transition

1 INTRODUCTION

The transition toward a green economy has emerged as one of the most pressing global development agendas of this decade. The United Nations Environment Programme defines a green economy as a system that simultaneously generates inclusive human well-being and substantially reduces environmental risks (UNEP, 2023). For developing nations, the pressure to adopt these principles presents a complex dilemma: on one hand, industrial growth is essential to absorb labor and increase national income, while on the other, increasingly stringent environmental standards demand comprehensive adaptation from business actors. Indonesia, as one of the largest economies in Southeast Asia, Indonesia's manufacturing value added accounted for about 19% of GDP, while manufacturing value added grew by 4.43% in 2024 (World Bank, 2024). These figures position Indonesia at a critical crossroads between the imperative of economic growth and the necessity of green transformation. The downstream industrialization policy momentum championed by the government since 2020 further reinforces the urgency of aligning domestic industrial capacity with global sustainability demands, creating dynamic space for the emergence of green-based innovation at the firm level.

Amid this dynamic, academic debate regarding the impact of environmental practices on firm performance continues. Part of the literature argues that environmental regulations add compliance burdens that suppress short-term profitability (Jaffe et al., 1995), while other literature contends that regulatory pressure stimulates innovation that ultimately enhances firm competitiveness (Porter and van der Linde, 1995). This complexity is compounded by the fact that most empirical studies have relied on proxy measures such as ISO certifications or CSR ratings, rather than granular and specific data on green technology implementation at the firm level (Phan, 2026). This gap is particularly evident in the Southeast Asian context, where cross-country studies exist but single-country empirical evidence grounded in detailed data remains highly limited. This study addresses that gap by utilizing the most recent 2023 World Bank Enterprise Survey (WBES) data, focusing specifically on the Indonesian context across both manufacturing and service sectors, to empirically test whether higher levels of Green Economy Practice (GEP) adoption increase the probability of product and process innovation at the firm level.

2 LITERATURE REVIEW

2.1 The Porter Hypothesis and Regulation-Driven Innovation

The first theoretical foundation underpinning this study is the Porter Hypothesis as articulated by Porter and van der Linde (1995). This hypothesis directly challenges the conventional assumption that environmental regulation invariably functions as an economic burden on firms. Porter and van der Linde (1995) argue that compliance pressure arising from carefully designed environmental standards compels firms to innovate efficiently, both in production processes and in the attributes of their outputs. Innovation generated through this pressure ultimately offsets and can even surpass the initial compliance costs borne by the firm, such that environmental regulation functions in the long run as a catalyst for competitiveness rather than merely a regulatory cost instrument. This argument laid the foundation for subsequent research that consistently found a positive correlation between the intensity of environmental regulation and the rate of innovation among firms operating in heavily regulated sectors (Jaffe et al., 1995).

2.2 Resource-Based View and Green Technological Capabilities

The second theoretical framework employed is the Resource-Based View (RBV) as originally formulated by Barney (1991) and subsequently extended by Hart (1995) specifically into the domain of firm-level environmental capabilities. Barney (1991) asserts that sustained competitive advantage can only derive from resources that are valuable, rare, inimitable, and non-substitutable. Within this perspective, investment in green physical capital such as clean production technologies or energy management systems constitutes a process of building firm-specific technological capabilities that competitors find difficult to replicate (Hart, 1995). These capabilities are cumulative in nature, meaning that the deeper a firm invests in green practices, the stronger its internal technological foundation becomes, which in turn continuously drives product and process innovation from within the organization itself.

2.3 Institutional Theory and Stakeholder Legitimacy

The third framework that complements the preceding two perspectives is Institutional Theory as developed by DiMaggio and Powell (1983). This theory holds that organizational behavior is not solely determined by rational economic calculation but is also shaped by isomorphic pressures emanating from the institutional environment surrounding the firm, encompassing government regulators, investors, business partners, and normative pressures from the broader public. DiMaggio and Powell (1983) identify three mechanisms of isomorphism, namely coercive, mimetic, and normative, which collectively drive firms toward conformity of practice within a given organizational field. In the context of green practices, these institutional pressures encourage firms to adopt environmentally friendly standards and technologies in order to secure legitimacy in the eyes of their stakeholders. The process of institutional adaptation demanded by these pressures inherently compels firms to undertake comprehensive overhauls of their internal systems and processes, which in turn catalyzes organizationally significant innovative change.

2.4 Environmental Management and Firm Performance

Empirically, the prior literature consistently documents a positive relationship between sound environmental management and improved firm performance across diverse economic contexts. Ambec and Lanoie (2008), in their comprehensive review of over 100 empirical studies, conclude that planned environmental initiatives can enhance firm revenues through broader market access while simultaneously reducing operational costs through energy and material efficiency, with estimated energy cost savings ranging from 10 to 30 percent among firms that fully adopt environmental management systems. Horbach (2008), using panel data from German firms, demonstrates that environmental regulatory pressure and the availability of internal knowledge resources are the primary determinants of eco-innovation, and that these two factors operate synergistically such that their combined effect exceeds the sum of their individual contributions. Nevertheless, the magnitude of environmental management's influence on firm performance varies considerably depending on the institutional context, sectoral characteristics, and the level of economic development of the country in question (Ambec and Lanoie, 2008; Horbach, 2008).

2.5 Green Economy Practices and Innovation in Emerging Markets

The most directly relevant empirical contribution to this study comes from Phan (2026), who specifically examines the relationship between the adoption of green economy practices and firm-level innovation across emerging markets. Using a broad cross-country dataset, Phan (2026) finds that the adoption of green economy practices is positively and significantly associated with both product and process innovation at the firm level, reporting odds ratios of 1.157 for product innovation and 1.194 for process innovation. These figures indicate that firms adopting green economy practices are approximately 1.16 times more likely to introduce product innovations and 1.19 times more likely to undertake process innovations relative to non-adopting firms, with effects that are stronger among firms operating in more conducive and stable regulatory environments. The study also highlights the continuing reliance of the literature on broad survey-based indicators of green practices, thereby opening opportunities for future research using more granular firm-level measures of green technology implementation.

2.6 Green Innovation and Firm Dynamics in Indonesia

In the specific context of Indonesia, several studies have examined the costs and benefits of green practice adoption among domestic business actors. Rizaldy et al. (2025) document that although the initial costs of green technology adoption are considerable, particularly for small and medium enterprises that on average commit initial investments equivalent to 15 to 20 percent of total operating assets, the long-term benefits in the form of energy efficiency gains and product quality improvements prove to exceed these initial investment burdens within a three to five year horizon. Yuliadhitya et al. (2021) add an export competitiveness dimension by demonstrating that Indonesian firms adopting environmentally friendly production standards enjoy broader international market access and record average export value growth approximately 12 percent higher than their non-adopting counterparts. Taken together, these findings reinforce the relevance of deeper empirical investigation into the relationship between green economy practices and firm-level innovation within Indonesia's industrial ecosystem, which carries distinctive institutional and structural characteristics.

3 METHODS

3.1 Research Data and Samples

This study utilizes cross-sectional, firm-level data from the 2023 World Bank Enterprise Survey (WBES) for Indonesia. The sample is restricted to respondents of the Version B questionnaire who completed the Green Economy Transition (GET) module, which covers the adoption of specific environmental practices across the manufacturing, retail, and other services sectors. This restriction yielded an initial sample of 1,512 firms. To ensure robust econometric estimation, listwise deletion was applied to remove observations with missing values in any of the analytical variables, a reduction primarily driven by firms lacking an assigned median survey weight (*wmedian_BR*). Consequently, the final analytical sample comprises 794 firm-level observations.

3.2 Model Description

Model 1 aims to investigate how the adoption of GEP affects product innovation:

$$\begin{aligned}\mathrm{PRODUCT}_i = \beta_0 + \beta_1 \mathrm{GEP\_Index}_i + \beta_2 \ln(\mathrm{SIZE}_i) \\ + \beta_3 \mathrm{GEP} \cdot \mathrm{SIZE}_i + \beta_4 \ln(\mathrm{AGE}_i) \\ + \beta_5 \mathrm{CFL}_i + \beta_6 \mathrm{FO}_i + \theta\, \mathrm{Sector\_FE}_i + \varepsilon_i \quad (1)\end{aligned}$$

Model 2 aims to investigate how the adoption of GEP affects process innovation:

$$\begin{aligned}\mathrm{PROCESS}_i = \beta_0 + \beta_1 \mathrm{GEP\_Index}_i + \beta_2 \ln(\mathrm{SIZE}_i) \\ + \beta_3 \mathrm{GEP} \cdot \mathrm{SIZE}_i + \beta_4 \ln(\mathrm{AGE}_i) \\ + \beta_5 \mathrm{CFL}_i + \beta_6 \mathrm{FO}_i + \theta\, \mathrm{Sector\_FE}_i + \varepsilon_i \quad (2)\end{aligned}$$

where i indexes individual firms. Survey sampling weights (*wmedian_BR*) are applied to ensure results are representative of the target population rather than reflecting the sample's oversampled strata. Heteroskedasticity-consistent standard errors (HC1) are used throughout. This study employs two binary innovation outcomes as dependent variables, one composite green economy index as the main independent variable, one interaction term, and a set of firm-level controls. Table 1 presents the full definition, data type, role, and operationalization notes for each variable.

Table 1: Definition and Operationalization of Research Variables

Variable	Data Type	Role	Notes
PRODUCT	Binary	Dependent	= 1 if firm introduced a new product or service; = 0 otherwise. Derived from survey item h1. "Don't Know" and "Refused" coded as missing.
PROCESS	Binary	Dependent	= 1 if firm introduced a new or significantly improved process; = 0 otherwise. Derived from survey item h5.
GEP_Index	Continuous (0–10)	Independent (Main)	Unweighted sum of binary adoption scores across 10 green economy practices (ge8a_BR to ge8j_BR). Each adopted practice scores 1.
GEP · SIZE	Continuous	Moderator	Interaction term. Tests whether firm size moderates the effect of GEP adoption on innovation. A negative coefficient indicates the GEP effect diminishes as firm size increases.
ln(SIZE)	Continuous	Control	Natural log of number of full-time workers (survey item l1). Log-transformed to reduce right skew.
ln(AGE)	Continuous	Control	Natural log of years since founding (2023 minus survey item b5). Log-transformed to reduce right skew.
CFL	Binary	Control	= 1 if firm holds a checking or savings account; = 0 otherwise (survey item k6). Proxy for internal liquidity.
FO	Continuous (0–100)	Control	Percentage of firm equity held by foreign shareholders (survey item b2b).
Sector FE	Binary	Control	Dummy variables from sector classification (survey item a0) covering manufacturing, retail, and other services. Manufacturing is the reference category.

Source: World Bank Enterprise Survey (WBES) Indonesia 2023, Version B.

3.3 Methodology

This study employs binary logistic regression (logit) as the primary estimation method, given that both dependent variables are binary outcomes. Logistic regression is appropriate in this context because it estimates the probability of a firm engaging in innovation as a function of a set of explanatory variables, with predicted values bounded between 0 and 1. The models are estimated via a Generalized Linear Model (GLM) framework with a binomial family and logit link function. To ensure that the estimates are representative of the target population rather than reflecting the oversampled strata of the WBES design, all models are estimated with survey sampling weights (*wmedian_BR*). Heteroskedasticity-consistent standard errors (HC1) are used throughout to ensure valid inference in the presence of non-constant error variance common in cross-sectional firm-level data.

To assess the robustness of the findings, both models are re-estimated using a probit link function. While the logit and probit models share the same underlying logic, they differ in their assumed distributional form. Logit assumes a logistic distribution, whereas probit assumes a standard normal distribution. The two specifications differ primarily in the weight assigned to observations in the tails of the distribution. Consistent results in terms of coefficient signs, statistical significance, and direction of effects across both link functions provide stronger evidence that the findings are not an artifact of the chosen functional form.

Since raw logistic regression coefficients are expressed in log-odds units and are not directly interpretable in terms of probability, this study additionally computes Average Marginal Effects (AME) for all variables. The AME measures the average change in the predicted probability of innovation associated with a one-unit increase in a given variable, evaluated at each observation's actual covariate values and averaged across all firms in the sample. This approach provides a more intuitive and policy-relevant quantification of the economic magnitude of each variable's effect on firm innovation outcomes.

4 RESULTS

4.1 Descriptive Statistics

Table 2 presents the descriptive statistics for the analytical sample of 794 firm-level observations used in the econometric estimation.

Table 2: Descriptive Statistics

Variable	N	Mean	Std. Dev.	Min	Max
PRODUCT	794	0.0466	0.2109	0.0000	1.0000
PROCESS	794	0.0214	0.1448	0.0000	1.0000
GEP_Index	794	2.1776	3.1635	0.0000	10.0000
CFL	794	0.7456	0.4358	0.0000	1.0000
Workers (level)	794	52.7695	141.0357	1.0000	2000.0000
FO	794	2.1134	12.6275	0.0000	100.0000
AGE (level)	794	19.0189	12.6908	1.0000	113.0000

Note: DEBT was excluded from summary statistics due to missing output in final cleaned sample.
Source: WBES Indonesia 2023, Version B.

On average, 4.66% of firms in the sample reported introducing a new product or service, while 2.14% reported introducing a new or significantly improved process. The mean GEP Index stands at 2.18, indicating that the average Indonesian firm in the sample has adopted approximately two of the ten available green practices, pointing to a relatively nascent stage of green economy engagement among Indonesian firms. The average firm has around 53 workers and has been operating for approximately 19 years. Around 74.56% of firms hold a checking or savings account, while foreign ownership is minimal on average at 2.11%.

The distribution of the GEP Index, displayed in Figure 1, reveals a pronounced right skew. A majority of firms, specifically 652 out of the full Version B sample, adopted zero green economy practices, while the frequency declines sharply at higher adoption levels. Only 80 firms adopted all ten practices. The high concentration of zero-adopters reflects the early-stage green transition among Indonesian firms and the limited enforcement of environmental standards in the domestic business environment.

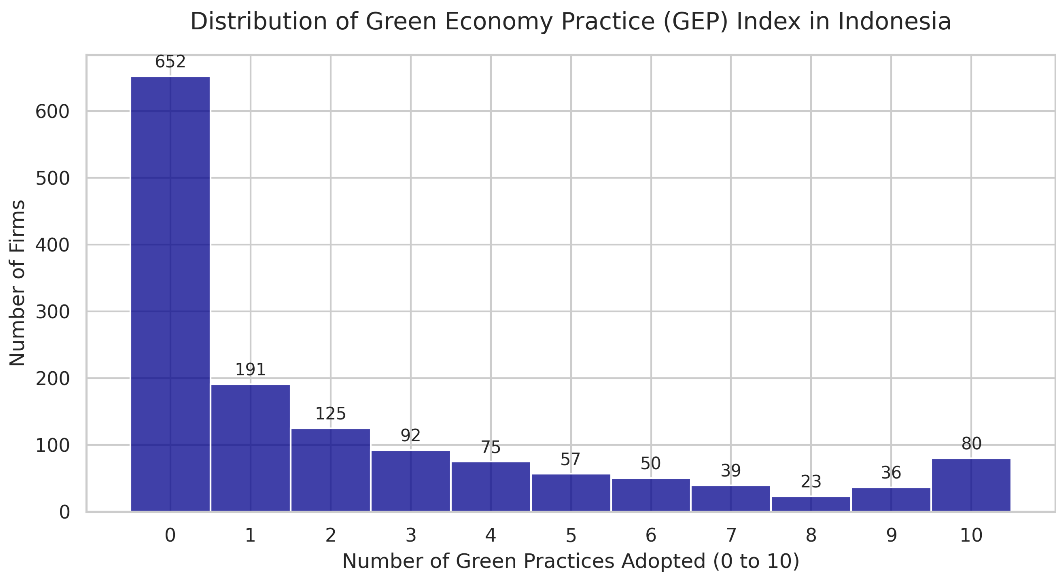

Figure 1: Distribution of GEP Index among Indonesian Firms

Source: WBES Indonesia 2023, Version B. Authors' own computation.

Figure 2 presents the adoption rates of each individual green practice. Machinery upgrades (33.9%) and waste minimization (33.6%) are the most commonly adopted practices, followed by lighting improvements (31.3%) and vehicle fleet upgrades (27.1%). In contrast, heating and cooling improvements (18.4%), on-site green energy generation (18.5%), and other pollution control measures (19.1%) are the least adopted. These patterns suggest that Indonesian firms tend to prioritize practices with more immediate operational returns, such as machinery and waste management, over energy generation and pollution control measures that may require higher upfront investment.

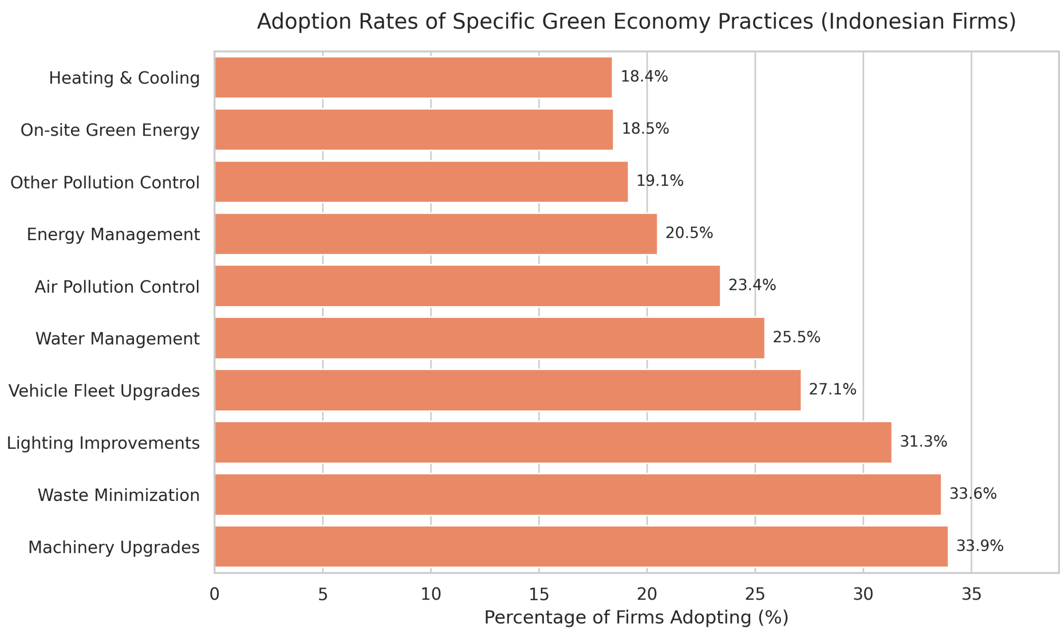

Figure 2: Adoption Rates of Individual Green Economy Practices (Indonesian Firms)

Source: WBES Indonesia 2023, Version B. Authors' own computation.

4.2 Logistic Regression Results

Table 3 presents the baseline logistic regression results for both innovation outcomes.

Table 3: Logistic Regression Results

	(1)	**(2)**
GEP_Index	0.464	1.132***
	(0.328)	(0.357)
GEP · SIZE	−0.104	−0.285**
	(0.110)	(0.124)
SIZE	−0.334	−0.101
	(0.621)	(0.349)
AGE	−0.273	2.061***
	(0.498)	(0.621)
CFL	0.740	0.887
	(1.110)	(1.000)
FO	−0.011	0.002
	(0.010)	(0.013)
Other Services	−0.707	2.260*
	(1.001)	(1.254)
Retail Services	−0.180	2.128
	(0.908)	(1.300)
Constant	−2.775	−13.891***
	(2.430)	(3.227)
Observations	794	794
Sector FE	Yes	Yes
Survey Weights	Yes	Yes

Note: Robust standard errors (HC1) in parentheses. Survey weights (*wmedian_BR*) applied. Sector fixed effects included with manufacturing as the reference category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Source: WBES Indonesia 2023, Version B.

Model 1 examines the determinants of product innovation. The GEP Index coefficient is positive ($\beta = 0.464$) but statistically insignificant ($p > 0.05$), indicating that green economy practice adoption does not generate a statistically detectable increase in the probability of product innovation in this sample. The interaction term GEP · SIZE is similarly insignificant. Among the control variables, none reach conventional significance levels, including firm age and cash flow, suggesting that product innovation in the Indonesian context may be driven by factors not captured in this model.

Model 2 examines the determinants of process innovation and yields substantively different results. The GEP Index is positive and highly significant, indicating that each additional green economy practice adopted is associated with markedly higher log-odds of process innovation. The interaction term GEP · SIZE is negative and significant ($\beta = -0.285$, $p < 0.05$), suggesting that the positive effect of GEP adoption on process innovation diminishes as firm size increases — larger firms appear to benefit less from additional green practices, possibly because they have already adopted baseline process improvements through non-green channels. Firm age ($\ln AGE$) is also positive and significant, indicating that older firms are more likely to engage in process innovation, consistent with the view that accumulated organizational capabilities facilitate process redesign.

4.3 Average Marginal Effects

Table 4 presents the Average Marginal Effects (AME) of the GEP Index and selected variables, calculated from the baseline logit models using survey weights.

Table 4: Average Marginal Effects (AME)

	(1)	**(2)**
GEP_Index	+1.55 pp	+7.46 pp***
GEP · SIZE	−0.35 pp	−1.88 pp**
SIZE	−1.12 pp	−0.67 pp
AGE	−0.91 pp	+13.59 pp***
CFL	+2.48 pp	+5.85 pp
FO	−0.04 pp	+0.01 pp

Note: AME computed at observed covariate values, weighted by *wmedian_BR*. Percentage points (pp) represent absolute probability changes. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Source: Authors' own computation from WBES Indonesia 2023, Version B.

For product innovation, the AME of the GEP Index is +1.55 percentage points but remains statistically insignificant, consistent with the logit results. For process innovation, adopting one additional green economy practice increases the probability of process innovation by approximately 7.46 percentage points. The interaction term GEP · SIZE yields an AME of −1.88 percentage points, confirming that the GEP effect on process innovation diminishes with firm size and is primarily operative among smaller firms. A one-unit increase in the natural log of firm age is associated with a 13.59 percentage point increase in the probability of process innovation, indicating that older firms are substantially more likely to engage in process innovation, consistent with the view that accumulated organizational experience facilitates process redesign.

4.4 Robustness Check

To verify that the baseline logit findings are not sensitive to the assumed distributional form, both models are re-estimated using a probit link function. Table 5 presents the probit results.

Table 5: Probit Model Estimations

	(1)	(2)
GEP_Index	0.236	0.480***
	(0.151)	(0.134)
GEP · SIZE	−0.056	−0.106**
	(0.047)	(0.045)
SIZE	−0.117	−0.031
	(0.241)	(0.150)
AGE	−0.137	0.881***
	(0.214)	(0.289)
CFL	0.314	0.284
	(0.417)	(0.395)
FO	−0.004	0.002
	(0.004)	(0.005)
Other Services	−0.464	0.812
	(0.373)	(0.499)
Retail Services	−0.123	0.804
	(0.376)	(0.499)
Constant	−1.518	−6.213***
	(0.967)	(1.347)
Observations	794	794
Sector FE	Yes	Yes
Survey Weights	Yes	Yes

Note: Robust standard errors (HC1) in parentheses. Survey weights (*wmedian_BR*) applied. Sector fixed effects included with manufacturing as the reference category. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.
Source: WBES Indonesia 2023, Version B.

For product innovation, the GEP Index remains positive but statistically insignificant ($\beta = 0.236$, $p > 0.1$), consistent with the null finding in the logit model. For process innovation, the GEP Index is again positive and highly significant, and the interaction term GEP · SIZE remains negative and significant. The directional consistency of all key coefficients across both specifications confirms that the findings are not artifacts of the logistic distributional assumption.

5 MODEL LIMITATIONS AND ROBUSTNESS

5.1 Limitations of the Study

While the findings offer meaningful empirical evidence on the relationship between GEP adoption and firm-level innovation in Indonesia, several limitations deserve honest acknowledgment. The cross-sectional nature of the WBES data means causal direction cannot be established with confidence, as firms that are already more innovative may simply have greater capacity to invest in green practices, rather than green practices driving innovation, and without longitudinal data this ambiguity cannot be resolved. The analytical sample of 794 firms, reduced from 1,512 due to listwise deletion driven by missing survey weights, may differ in unobserved ways from the broader population of Indonesian firms, particularly smaller and informal enterprises that tend to be underrepresented after weighting procedures are applied. Finally, the models rely solely on firm-level observables from the WBES questionnaire, leaving out factors that plausibly shape innovation outcomes, including local regulatory enforcement intensity, R&D infrastructure access, and supply chain dynamics, while the coarse three-sector fixed effects may obscure important within-sector heterogeneity that a finer industry classification would reveal.

5.2 Robustness of the Findings

To verify that the baseline logit findings are not driven by the choice of distributional assumption, both models were re-estimated using a probit link function. The direction, sign, and significance pattern of all key coefficients remain consistent across both specifications, providing confidence that the results are not artifacts of the logistic functional form. Throughout all estimations, heteroskedasticity-consistent standard errors (HC1) were applied to ensure valid inference in the presence of non-constant error variance common in cross-sectional firm-level data, and survey sampling weights (*wmedian_BR*) were used to ensure that the estimates reflect population-level patterns rather than the composition of the raw oversampled strata.

6 CONCLUSION AND RECOMMENDATION

6.1 Conclusion

This study examined whether Green Economy Practice (GEP) adoption increases the likelihood of product and process innovation among Indonesian firms using firm-level data from the World Bank Enterprise Survey 2023. For product innovation, GEP adoption yields no statistically significant effect across all specifications, suggesting that converting green capabilities into new commercial offerings requires a broader enabling environment, including stronger market demand, deeper R&D infrastructure, and university-industry linkages, that most Indonesian firms do not yet have access to. For process innovation, the effect is both statistically robust and economically meaningful: each additional green practice adopted raises the probability of process innovation by approximately 7.46 percentage points against a baseline mean of just 2.14%, implying a very large proportional shift that is concentrated among smaller firms and amplified by organizational age, consistent with the view that green investment forces internal system rebuilding and that accumulated experience is what allows firms to translate that rebuilding into genuine operational change. The finding that the average Indonesian firm has adopted only about two of ten possible green practices, with a majority adopting none at all, underscores how nascent this transition remains and how much of the potential innovation dividend has yet to be unlocked.

6.2 Recommendation

Policymakers should move beyond simply adding new environmental regulations and instead focus on making existing standards more credible, consistent, and enforceable, since it is the stability of regulatory pressure rather than its volume that the Porter Hypothesis identifies as the true catalyst for innovation responses. This should be paired with targeted financial incentives, such as preferential credit access or procurement advantages designed specifically for smaller firms, which the data shows stand to gain the most from green adoption but face the steepest upfront cost barriers. At the same time, closing the gap toward product innovation requires investments that go beyond the firm level entirely, including applied clean technology research, stronger linkages between firms and research institutions, and demand-side programs that build viable domestic markets for green products, because without these complementary conditions the process-level gains documented here risk becoming the ceiling of Indonesia's green innovation potential rather than its foundation.

REFERENCES

Ambec, S., & Lanoie, P. (2008). Does it pay to be green? A systematic overview. *Academy of Management Perspectives*, 22(4), 45–62.

Barney, J. (1991). Firm resources and sustained competitive advantage. *Journal of Management*, 17(1), 99–120.

DiMaggio, P. J., & Powell, W. W. (1983). The iron cage revisited: Institutional isomorphism and collective rationality in organizations. *American Sociological Review*, 48(2), 147–160.

Hart, S. L. (1995). A natural-resource-based view of the firm. *Academy of Management Review*, 20(4), 986–1014.

Horbach, J. (2008). Determinants of environmental innovation: New evidence from German panel data sources. *Research Policy*, 37(1), 163–173.

Jaffe, A. B., Peterson, S. R., Portney, P. R., & Stavins, R. N. (1995). Environmental regulation and the competitiveness of U.S. manufacturing: What does the evidence tell us? *Journal of Economic Literature*, 33(1), 132–163.

Phan, Q. (2026). The impact of green economy practices on firm innovation using data from the World Bank Enterprise Surveys. *Discover Sustainability.*

Porter, M. E., & van der Linde, C. (1995). Toward a new conception of the environment-competitiveness relationship. *Journal of Economic Perspectives*, 9(4), 97–118.

Rizaldy, H., Moslehpour, M., Gupta, V. et al. (2025). Waking up to the costs of green technology adoption integrating economic and environmental priorities. *Discover Sustainability*, 6, 398.

United Nations Environment Programme. (2023). *Global environment outlook: Healthy planet, healthy people.* UNEP.

World Bank. (2024). *Indonesia economic prospects: Strengthening competitiveness.* World Bank Group.

Yuliadhitya S, F., Retnandari, N. D., & Dr., M.Si. (2021). *Innovation and export performance: A study of the Indonesian manufacturing exporters.* Universitas Gadjah Mada.